

Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects*

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Abstract

To estimate the dynamic effects of an absorbing treatment, researchers often use two-way fixed effects regressions that include leads and lags of the treatment. We show that in settings with variation in treatment timing across units, the coefficient on a given lead or lag can be contaminated by effects from other periods, and apparent pretrends can arise solely from treatment effects heterogeneity. We propose an alternative estimator that is free of contamination, and illustrate the relative shortcomings of two-way fixed effects regressions with leads and lags through an empirical application.

Keywords: difference-in-differences, two-way fixed effects, pretrend test

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following differences. First, in the setting where there is no never-treated group, our method uses the last cohort to be treated as a control group, whereas Callaway and Sant’Anna (2020a) use the set of not-yet-treated cohorts. Our method and theirs thus rely on different, but non-nested parallel trends assumptions. Second, our estimation method can be cast as a regression specification and thus may be more familiar to applied researchers. However, a third difference is that the procedure of Callaway and Sant’Anna (2020a) allows for conditioning on time-varying covariates. de Chaisemartin and D’Haultfœuille (2020) and Goodman-Bacon (2018) respectively propose alternative estimators and diagnostic tools for estimation of causal effects in staggered settings, but do not consider the estimation of the dynamic path of treatment effects as we do.

2 Event studies design

In this section we first formalize the “event studies design”. As discussed in Section 2.3, based on how this term is deployed in the empirical literature, an *event study* design is a *staggered adoption* design where units are treated at different times, and there may or may not be never treated units. It also nests a *difference-in-differences* design, where units are either first treated at time t_0 or never treated.

Specifically, we consider a setting with a random sample of N units observed over $T + 1$ time periods, where T is fixed. For each $i \in \{0, \dots, N\}$ and $t \in \{0, \dots, T\}$, we observe the outcome $Y_{i,t}$ and treatment status $D_{i,t} \in \{0, 1\}$: $D_{i,t} = 1$ if i is treated in period t and $D_{i,t} = 0$ if i is not treated in period t . Throughout we assume that the observations $\{Y_{i,t}, D_{i,t}\}_{t=0}^T$ are independent and identically distributed (i.i.d.).

In the general case of event studies we focus on an *absorbing* treatment such that the treatment status over time is a non-decreasing sequence of zeros and then ones, i.e. $D_{i,s} \leq D_{i,t}$ for $s < t$. We can thus uniquely characterize a treatment path by the time period of the initial treatment, denoted with $E_i = \min \{t : D_{i,t} = 1\}$. If unit i is never treated i.e. $D_{i,t} = 0$ for all t , we set $E_i = \infty$. Based on when they first receive the treatment, we can also uniquely categorize units into disjoint cohorts e for $e \in \{0, \dots, T, \infty\}$, where units in cohort e are first treated at the same time $\{i : E_i = e\}$.

We define $Y_{i,t}^e$ to be the potential outcome in period t when unit i is first treated in time period e . We define $Y_{i,t}^\infty$ to be the potential outcome if unit i never receives the treatment, which we call the “baseline outcome”. Since the timing of the initial treatment uniquely characterizes one’s treatment path, we can

treatment is

$$CATT_{e,\ell} = E[Y_{i,e+\ell} - Y_{i,e+\ell}^{\infty} \mid E_i = e]. \quad (3)$$

Each $CATT_{e,\ell}$ represents the average treatment effect ℓ periods from the initial treatment for the cohort of units first treated at time e . We shift from calendar time index t to relative period index ℓ which denotes the periods since treatment; for cohort e , ℓ ranges from $-e$ to $T - e$ because we observe at most e periods before the initial treatment and $T - e$ periods after the initial treatment. Relative periods allow us to compare cohorts while holding their exposure to the treatment constant.

2.2 Identifying assumptions

With the above definitions, we formalize three potential identifying assumptions for outcomes of interest in our event study design. The first assumption is a generalized form of a parallel trends assumption. The second assumption requires no anticipation of the treatment. The third assumption imposes no variation across cohorts. For each assumption, we first discuss its meaning and then compare it with similar assumptions made in the literature interpreting two-way fixed effects regressions. Later in Section 3 we interpret the relative period coefficients μ_{ℓ} from two-way fixed effects regressions under different combinations of these assumptions.

Assumption 1. (Parallel trends in baseline outcomes.) *For all $s \neq t$, the $E[Y_{i,t}^{\infty} - Y_{i,s}^{\infty} \mid E_i = e]$ is the same for all $e \in \text{supp}(E_i)$.*

If an application includes never-treated units so that $\infty \in \text{supp}(E_i)$, we need to especially consider whether these never-treated units satisfy the parallel trends assumption. Never-treated units are likely to differ from ever-treated units in many ways, and may not share the same evolution of baseline outcomes. If the never-treated units are unlikely to satisfy the parallel trends assumption, then we should exclude them from the estimation to avoid violation of this assumption.

While common in the applied literature, the parallel trends assumption is strong and oftentimes violated. For example, Ashenfelter (1978) documented that participants in job training programs experience a decline in earnings prior to the training period (Ashenfelter's dip). The timing of job training is dependent on the evolution of individual's baseline earnings, and this scenario therefore does not satisfy the parallel trends assumption. Proposition 1 is the only result in this paper derived without this assumption, but there is an

active literature studying inference under violations of the parallel trends assumption e.g. Rambachan and Roth (2020).

Our parallel trends assumption coincides with that of de Chaisemartin and D’Haultfœuille (2020). One could substitute this assumption with a different identifying assumption that baseline outcomes are mean independent of E_i i.e. at each t , $E[Y_{i,t}^\infty | E_i = e]$ is the same for all $e \in \text{supp}(E_i)$ and in particular is equal to $E[Y_{i,t}^\infty]$. This stronger assumption is plausible when the timing of treatment is indeed randomized, which is the assumption used by Athey and Imbens (2018). By taking the “fully dynamic” specification as their DGP, Borusyak and Jaravel (2017) implicitly assume this version of a parallel trends assumption. Callaway and Sant’Anna (2020a) propose a weaker version that is conditional on covariates. Finally, for a particular estimand, Goodman-Bacon (2018) identifies a weaker version that only requires a weighted average of $E[Y_{i,t}^\infty - Y_{i,s}^\infty | E_i = e]$ (averaged across cohorts) to be zero.

Assumption 2. (No anticipatory behavior prior to treatment.) *There is no treatment effect in pre-treatment periods i.e. $E[Y_{i,e+\ell}^e - Y_{i,e+\ell}^\infty | E_i = e] = 0$ for all $e \in \text{supp}(E_i)$ and all $\ell < 0$.*

Assumption 2 requires potential outcomes in any ℓ periods before treatment to be equal to the baseline outcome on average as in Malani and Reif (2015) and Botosaru and Gutierrez (2018). This is most plausible if the full treatment paths are not known to units. If they have private knowledge of the future treatment path they may change their behavior in anticipation and thus the potential outcome prior to treatment may not represent baseline outcomes. For example, Hendren (2017) shows that knowledge of future job loss leads to decreases in consumption. If the periods with anticipation behavior are known, then we may consider an alternative version of Assumption 2, which holds for pre-periods in a subset of pre-treatment periods. Depending on the application, it may still be plausible to assume no anticipation until K periods before the treatment.

The no anticipation assumption proposed by Athey and Imbens (2018) is a deterministic condition which stipulates that $Y_{i,e+\ell}^e = Y_{i,e+\ell}^\infty$ for all units i and e and $\ell < 0$. By taking the “fully dynamic” specification as their DGP, Borusyak and Jaravel (2017) allow anticipation by including pre-trends indicators in the DGP. Callaway and Sant’Anna (2020a) and Goodman-Bacon (2018) implicitly assume no anticipation by using observed outcomes in time periods before the initial treatment as the untreated potential outcomes.

Assumption 3. (Treatment effect homogeneity.) *For each relative period ℓ , $CATT_{e,\ell}$ does not depend on cohort e and is equal to ATT_ℓ .*

Assumption 3 requires that each cohort experiences the same path of treatment effects. Treatment effects need to be the same across cohorts in every relative period for homogeneity to hold, whereas for heterogeneity to occur, treatment effects only need to differ across cohorts in one relative period. The assumption of treatment effect homogeneity is therefore strong, and in Section 3.4.1, we describe how it can be violated in applied settings.

Our notion of treatment effect homogeneity does not preclude dynamic treatment effects; it only imposes that cohorts share the same path of treatment effects. The related literature sometimes formulates restrictions on the dynamics of treatment effects as another notion of treatment effect homogeneity. Athey and Imbens (2018) propose an assumption that “restricts the heterogeneity of the treatment effects over time,” which implies $CATT_{e,\ell}$ can vary over e but not over ℓ . Borusyak and Jaravel (2017) refer to one type of treatment effects heterogeneity as “only across the time horizon,” which implies $CATT_{e,\ell}$ can vary over ℓ but not over e . Callaway and Sant’Anna (2020a) allow for “arbitrary treatment effect heterogeneity” when $CATT_{e,\ell}$ varies across cohorts and over time. Similarly, de Chaisemartin and D’Haultfœuille (2020) describe treatment effects that may be “heterogeneous across groups and over time periods.” Goodman-Bacon (2018) allow heterogeneous effects to either “vary across units but not over time” or “vary over time but not across units.” The literature has not converged on a single notion of treatment effects heterogeneity with time-varying treatment. Since researchers are interested in dynamic treatment effects when using a “dynamic” specification, we do not restrict the path of treatment effects but rather use “heterogeneity” to describe variation across cohorts only.

2.3 Relevance in the applied literature

To gauge the empirical relevance of our results, we survey the estimation methods used by the twelve papers collected by Roth (2019) from three leading economics journals that contain the phrase “event study” in their main text.¹ From this sample of applied papers, we learn what specifications empirical researchers are actually using when estimating two-way fixed effects regressions. Four papers in this sample consider the simple setting where units either receive their first treatment at the same time or never receive the treatment. The other eight papers in this sample consider the more complex setting where treated units receive their

¹We follow the selection criteria in Roth (2019): the original sample consists of 70 total papers, but is further constrained to these twelve papers with publicly available data and code. The data and code are used to determine exactly the specification estimated in these papers.

for all ℓ and all t . We can represent the relative period bin indicator as

$$\mathbf{1}\{t - E_i \in g\} = \sum_{\ell \in g} \mathbf{1}\{t - E_i = \ell\} = \sum_{\ell \in g} D_{i,t}^\ell. \quad (5)$$

Static specification. For a “static” specification $\mathcal{G}$ contains a single element equal to $g = [0, T]$. The indicator $\mathbf{1}\{t - E_i \in g\}$ is equivalent to an indicator for whether unit i has received its initial treatment by t : $\mathbf{1}\{E_i \leq t\}$. The “static” specification thus takes the following form

$$Y_{i,t} = \alpha_i + \lambda_t + \mu_g \sum_{\ell \geq 0} D_{i,t}^\ell + v_{i,t} \quad (6)$$

and the corresponding set of excluded relative periods is $g^{excl} = [-T, -1]$.

Dynamic specification. “Dynamic” specifications encompass any specifications of $\mathcal{G}$ where $\mathcal{G}$ contains more than one element, thus treatment effects are allowed to vary over time non-parametrically. In its most flexible form, a “fully dynamic” specification takes the following form

$$Y_{i,t} = \alpha_i + \lambda_t + \sum_{\ell=-K}^{-2} \mu_\ell D_{i,t}^\ell + \sum_{\ell=0}^L \mu_\ell D_{i,t}^\ell + v_{i,t} \quad (7)$$

and the corresponding set of excluded relative periods is $g^{excl} = \{-T, \dots, -K-1, -1, L+1, \dots, T\}$.

Excluding some relative periods from the “fully dynamic” specification is necessary to avoid multi-collinearity, either among the relative period indicators $D_{i,t}^\ell$, or with the unit and time fixed effects. For example, when there are no never-treated units i.e. $\infty \notin \text{supp}(E_i)$ but with a panel balanced in calendar time, we need to exclude at least two relative period indicators in $\mathcal{G}$. These collinearities are discussed by Borusyak and Jaravel (2017): one multi-collinearity comes from the relative period indicators summing to one for every unit $\sum_{\ell \in [-T, T]} D_{i,t}^\ell = 1$, and the other multi-collinearity comes from the linear relationship between two-way fixed effects and the relative period indicators, namely $t - E_i = \ell$.

Excluding relative periods close to the initial treatment is common in practice. Normalizing relative to the period prior to treatment is the most common - six out of the eight papers we survey do so, as reflected in the above specification where we drop $D_{i,t}^{-1}$. The remaining two papers exclude $D_{i,t}^0$.

Excluding distant relative periods is however less common (only one of the eight papers we survey does so). Instead researchers “bin” or “trim” distant relative periods. For “binning”, researchers bin distant

the closed-form expressions for the weights using the classical “omitted variables bias formula” in Section 3.6. We defer the formal derivation to the Appendix. Here we mention the following four properties of the weights $\omega_{e,\ell}^g$.

- For relative periods of μ_g ’s own bin i.e. $\ell \in g$, their associated weights as displayed in (9) sum to one $\sum_{\ell \in g} \sum_e \omega_{e,\ell}^g = 1$.
- For relative periods belonging to some other bin included in (4) i.e. $\ell \in g'$ for $g' \neq g$ and $g' \in \mathcal{G}$, their associated weights as displayed in (10) sum to zero $\sum_{\ell \in g'} \sum_e \omega_{e,\ell}^g = 0$ for each bin g' .
- For relative periods not included in $\mathcal{G}$, their associated weights as displayed in (11) sum to negative one $\sum_{\ell \in g^{excl}} \sum_e \omega_{e,\ell}^g = -1$.
- If there are never-treated units i.e. $\infty \in \text{supp}(E_i)$, we have $\omega_{\infty,\ell}^g = 0$ for all g and ℓ .

We can easily estimate the weights $\omega_{e,\ell}^g$ for any given specification of $\mathcal{G}$ using the following auxiliary regression:

$$D_{i,t}^\ell \cdot \mathbf{1}\{E_i = e\} = \alpha_i + \lambda_t + \sum_{g \in \mathcal{G}} \omega_{e,\ell}^g \mathbf{1}\{t - E_i \in g\} + v_{i,t} \quad (12)$$

which regresses $D_{i,t}^\ell \cdot \mathbf{1}\{E_i = e\}$ on all bin indicators included in regression (4) and two-way fixed effects.

All of the above properties can be extended to a case where covariates are added to regression (4) by partialling out the covariates before proceeding. In other words, the terms in parentheses in (9), (10) and (11) would be replaced by terms for which the covariates are partialled out. The weights can be estimated by controlling for covariates in regression (12) the same way as they are controlled for in the original regression.

However, covariates complicate the interpretations of μ_g in terms of $CATT_{e,\ell}$ as we describe below in Proposition 2-4. Depending on how covariates are controlled for in regression (4), we may need an additional assumption that the counterfactual trends are linear in the time-varying covariates $X_{i,t}$. We leave a full investigation of the introduction of covariates for future work.

3.3 Interpreting the coefficients under parallel trends assumption only

Proposition 2. *Under Assumption 1 (parallel trends) only, the population regression coefficient on the indicator for relative period bin g is a linear combination of $CATT_{e,\ell \in g}$ as well as $CATT_{e,\ell'}$ from other*

each $CATT_{e,\ell}$ using an interacted two-way fixed effects regression, then estimate the weight $Pr\{E_i = e \mid E_i \in [-\ell, T - \ell]\}$ using their sample analogs. In the final step, we average over the cohort-specific estimates associated with relative period ℓ . This method has a similar flavor as the method proposed by Gibbons et al. (2019). They first use an interacted model to estimate the treatment effect for each fixed effect group; the resulting group-specific estimates are averaged to provide the ATE. Their method improves fixed effects regressions in a cross-sectional setting, and our method builds on theirs by improving two-way fixed effects regressions in a panel setting. We therefore follow their terminology in calling our alternative estimator an “interaction-weighted” estimator.

4.1 Interaction-weighted estimator

We describe the estimation procedure in three steps (with more detailed definitions stated in Definition 4 of Online Appendix B).

Step 1. We estimate $CATT_{e,\ell}$ using a linear two-way fixed effects specification that interacts relative period indicators with cohort indicators, excluding indicators for cohorts from some set C :

$$Y_{i,t} = \alpha_i + \lambda_t + \sum_{e \notin C} \sum_{\ell \neq -1} \delta_{e,\ell} (\mathbf{1}\{E_i = e\} \cdot D_{i,t}^\ell) + \epsilon_{i,t}. \quad (26)$$

The exact specification depends on the cohort shares for a given application. If there is a never-treated cohort, i.e. $\infty \in \text{supp}\{E_i\}$, then we may set $C = \{\infty\}$ and estimate regression (26) on all observations. If there are no never-treated units, i.e. $\infty \notin \text{supp}\{E_i\}$, then we may set $C = \{\max\{E_i\}\}$, i.e. the latest-treated cohort and estimate regression (26) on observations from $t = 0, \dots, \max\{E_i\} - 1$. Lastly, if there is a cohort that is always treated, i.e. $0 \in \text{supp}\{E_i\}$, then we need to exclude this cohort from estimation.

The coefficient estimator $\widehat{\delta}_{e,\ell}$ from regression (26) is a DID estimator for $CATT_{e,\ell}$ with particular choices of pre-periods and control cohorts. As DID is likely a familiar estimator for applied researchers, we separate the more in-depth discussion of its definition, choices of pre-periods, and choices of control cohorts in Section 4.2.

Step 2. We estimate the weights $Pr\{E_i = e \mid E_i \in [-\ell, T - \ell]\}$ by sample shares of each cohort in the relevant period(s) $\ell \in g$.

Step 3. To form our IW estimator, we take a weighted average of estimates for $CATT_{e,\ell}$ from Step 1

with weight estimates from step 2. More formally, the IW estimator is

$$\widehat{v}_g = \frac{1}{|g|} \sum_{\ell \in g} \sum_e \widehat{\delta}_{e,\ell} \widehat{Pr}\{E_i = e \mid E_i \in [-\ell, T - \ell]\} \quad (27)$$

where $\widehat{\delta}_{e,\ell}$ is returned from step 1 and $\widehat{Pr}\{E_i = e \mid E_i \in [-\ell, T - \ell]\}$ is the estimated weight returned from step 2. We normalize the weights further by the size of g . If g is a singleton, then its size is $|g| = 1$.

Validity of the IW estimator. Under the parallel trends and no anticipation assumptions the coefficient estimator $\widehat{\delta}_{e,\ell}$ from regression (26) is a consistent estimator for $CATT_{e,\ell}$. The sample shares of each cohort are also consistent estimators for the population shares. Thus, the IW estimator is consistent for a weighted average of $CATT_{e,\ell}$ with weights equal to the share of each cohort in the relevant period(s).

With a few standard assumptions (which we present as Assumption 4 in Online Appendix B) on regression (26), we can show that each IW estimator is asymptotically normal and derive its asymptotic variance. The large sample approximation allows us to estimate the variance of IW estimators directly without relying on bootstrapping as in Callaway and Sant'Anna (2020a). However, we only construct pointwise confidence interval valid for a given IW estimator $\widehat{v}_g$. The bootstrap-based inference by Callaway and Sant'Anna (2020a) constructs simultaneous confidence intervals that are valid for the entire path of $\widehat{v}_g$.

4.2 Difference-in-differences estimator for $CATT_{e,\ell}$

Definition 2. Assume cohort e is non-empty i.e. $\sum_{i=1}^N \mathbf{1}\{E_i = e\} > 0$. Assume there exists some pre-period $s < e$ and some set of control cohorts $C \subseteq \{c : e + \ell < c \leq T\}$ that are non-empty i.e. $\sum_{i=1}^N \mathbf{1}\{E_i \in C\} > 0$. Using the notion $\mathbb{E}_N$ to abbreviate the symbol $\frac{1}{N} \sum_{i=1}^N$, the DID estimator with pre-period s and control cohorts C estimates $CATT_{e,\ell}$ as

$$\widehat{\delta}_{e,\ell} = \frac{\mathbb{E}_N[(Y_{i,e+\ell} - Y_{i,s}) \cdot \mathbf{1}\{E_i = e\}]}{\mathbb{E}_N[\mathbf{1}\{E_i = e\}]} - \frac{\mathbb{E}_N[(Y_{i,e+\ell} - Y_{i,s}) \cdot \mathbf{1}\{E_i \in C\}]}{\mathbb{E}_N[\mathbf{1}\{E_i \in C\}]} \quad (28)$$

The assumption of non-empty cohort e , existence of pre-period and non-empty control cohorts makes the DID estimator well-defined. For example, DID estimators for cohort 0 are not well-defined because a pre-period does not exist for this cohort, which is why we exclude them from estimating regression (26) in Step 1.

Using the above definition, in regression (26) from Step 1 of our proposed method, the coefficient

estimator $\widehat{\delta}_{e,\ell}$ is a DID estimator for $CATT_{e,\ell}$ with pre-period $s = e - 1$ (because we exclude relative period $\ell = -1$) and some choice of control cohorts C . If there is a never-treated cohort i.e. $\infty \in \text{supp}\{E_i\}$, then we set the control cohort to be never-treated units $C = \{\infty\}$. If there are no never-treated units, i.e. $\infty \notin \text{supp}\{E_i\}$, then we set the control cohort $C = \{\max\{E_i\}\}$, i.e. the latest-treated cohort. Among all possible pre-periods, for regression (26) from step 1 of our proposed method we choose $s = e - 1$ and $C = \{\infty\}$ with never-treated units (or $\{\max\{E_i\}\}$ without never-treated units) because the resulting specification is a natural extension of the common specifications of two-way fixed effects regression.

Note that without never-treated units we need to drop time periods $t \geq \max\{E_i\}$ from estimating regression (26) because every unit will be treated in these periods. The DID estimators for $CATT_{e,\ell}$ for $e + \ell \geq \max\{E_i\}$ are thus not well-defined as the control cohort is empty $C = \emptyset$. For example, when there are just two cohorts with $E_i \in \{1, T\}$, one treated at $t = 1$ and the other treated in the last period, we then need to omit interaction terms involving the latest-treated cohorts as well as dropping observations from the last period $t = T$ from estimation.

Under some assumptions, this DID estimator $\widehat{\delta}_{e,\ell}$ is an unbiased and consistent estimator for $CATT_{e,\ell}$, a fact that we build on in deriving the probability limit of the IW estimator. We state this in the following proposition.

Proposition 5. *If Assumptions 1 and 2 hold, then the DID estimator using any pre-period $s < e$ and non-empty control cohorts C is an unbiased and consistent estimator for $CATT_{e,\ell}$.*

It is possible to relax the parallel trends assumption to allow the timing of treatment to depend on covariates. One can estimate $CATT_{e,\ell}$ consistently based on the inverse propensity score reweighted estimator proposed by Abadie (2005) and Callaway and Sant’Anna (2020a), the outcome regression approach by Heckman et al. (1997), and the doubly robust estimator recently proposed by Sant’Anna and Zhao (2020). The resulting estimates $\widehat{\delta}_{e,\ell}$ can then be plugged into step 3 to form our IW estimator. In particular, without covariates and for the case with never treated units, our approach coincides with Callaway and Sant’Anna (2020a). Therefore one can use the did R package developed by Callaway and Sant’Anna (2020b) to form the IW estimator.

The choice of the pre-period s and the control cohorts C depends on the trade-off between relaxing Assumption 1 or Assumption 2. If Assumption 2 is likely to hold, we can choose any pre-period $s < e$ and include not-yet-treated units in control cohorts $C = \{c : c > e + \ell\}$ as in Callaway and Sant’Anna (2020a). This